

Spring 2025

Introduction to Artificial Intelligence

Homework 5: Generative AI

Due Date: 2025/06/13 17:00

Introduction

Generative AI (Gen AI) has become a trending topic worldwide, especially after the release of ChatGPT. Gen AI can generate various types of content, including text, images, audio, and video. Recently, in the field of image generation, a novel technique called the Diffusion Model has gained significant attention, which this assignment focuses on. In this task, you will implement the diffusion process and train your own diffusion model from scratch. Additionally, you will generate a special-effect image using a pre-trained text-to-image diffusion model. This special effect requires an additional denoising operation without requiring further training or fine-tuning.

The goal of this programming assignment is to:

- 1) Understand the architecture and design of U-Net.
- 2) Implement a diffusion model and train it from scratch.
- 3) Learn how to evaluate generative models (e.g., FID score).
- 4) Utilize existing (diffusion) models available on Hugging Face.
- 5) Explore a training-free method for diffusion.

Requirements

1. Please modify the code in [source code](#) between **# Begin your code** and **# End your code**. If you modify the other code (e.g., Add HorizontalFlip() for data augmentation), please specify in the corresponding section or appendix.
2. This assignment requires **high-level** hardware resources (GPU), we suggest running your code on [Colab](#) or [Kaggle](#). We recommend using **Kaggle** since it allows a free account to access **T4 GPU up to 30 hours** weekly.

Codes on Kaggle Notebook: [Part1](#) | [Part2](#) | [FID&AFD Evaluation](#)

3. **Do not use** extra data, pre-trained weights, or external packages. Submitting the dataset as output may result in **zero points**. Ensure your code is reproducible; significant differences in our results will **reduce your score**.

Implementation (70%)

Part 1: Anime Face Generation (45%)

Learning Objectives: Why do we need to learn this ?

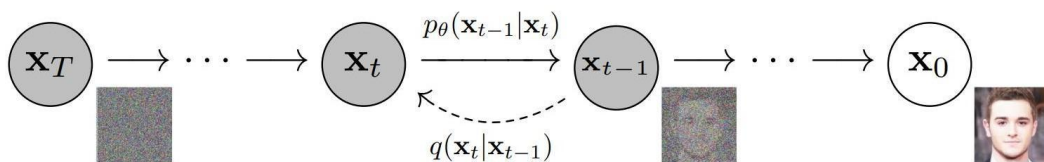
Training

1. Comprehend the Fundamentals of Diffusion Models (e.g., forward)
2. Interpret Key Mathematical Notations and Parameters (e.g., α_t)
3. Evaluate Generative Model Performance (e.g., FID score)
4. Explore Techniques to Enhance Model Performance (e.g., timesteps)

It's essential if you want to dive into related papers or this field!

Part 1-1: Denoising Process (15%)

- In a diffusion model, there are two distinct processes. The first is the **forward** process, which gradually adds noise to an image over time, with the noise sampled from a Gaussian distribution. The second is the **backward** process, where the model predicts the added noise from the image at time t .



(The image depicts the DDPM process, but we use the DDIM formula instead.)

Initialization Given β_t ; $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$

Forward $q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$

Backward $p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_t(x_t), \sigma_t^2 I)$

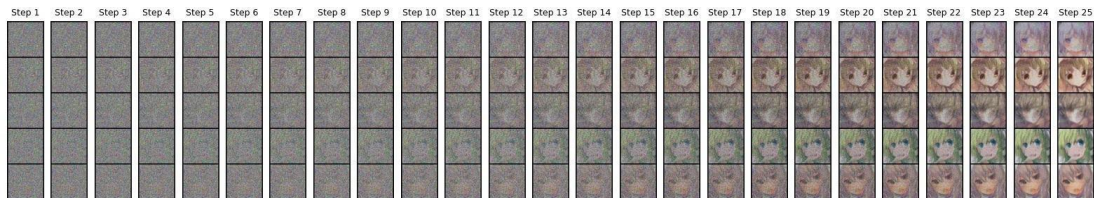
$$\mu_t(x_t) = \sqrt{\bar{\alpha}_{t-1}}\hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \epsilon_\theta(x_t, t)$$

$$\sigma_t = \eta \cdot \sqrt{\frac{(1 - \bar{\alpha}_{t-1})}{(1 - \bar{\alpha}_t)}} \cdot \sqrt{1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}}$$

In this section, you are required to complete the denoising process according to the above formula. The formula includes **initialization**, **forward**, and **backward**, which correspond to `init()`, `add_noise_forward()`, and `denoise_backward()` functions in a class named **GaussianDiffusion**.

Part 1-2: Result Visualization (10%)

- To keep track of the capabilities of the current diffusion model, it's helpful to visualize some intermediate results for analysis. First, you need to display the [denoising progress](#) for five different images. By plotting the intermediate steps, we can analyze efficiency and stability, and identify potential issues like over-smoothing or artifacts. Second, you need to plot the [loss curve](#) for each training loop. The loss curve can help us identify whether the model converged.



Part 1-3: Evaluation Baseline (20%)

- In image generation tasks, [FID scores](#) are commonly used to evaluate the quality of generated images by measuring their similarity to real images. A lower FID score indicates better quality. In our task, we also apply [Anime Face Detection \(AFD\)](#) to help determine whether an image contains a recognizable anime-style face. Using a pre-trained model, AFD ensures that generated faces are valid and coherent. The grading criteria are as follows:

Standard	FID Score	AFD Rate	Grade
Default	≤ 160.0	≥ 0.60	0%
Simple	≤ 120.0	≥ 0.70	5%
Normal	≤ 100.0	≥ 0.80	5%
Medium	≤ 90.0	≥ 0.85	5%
Hard	≤ 80.0	≥ 0.90	3%
Boss	≤ 70.0	≥ 0.95	2%

(You must satisfy both the FID score and AFD rate to receive the full score.)
(e.g. FID 93.4, AFD 0.88 ► Simple✓ Normal✓ Medium✗ Hard✗ Boss✗)

Since the original code doesn't have great performance, you need to make some [adjustments](#) to improve it. Here are some strategies you might want to attempt: **(1) Hyperparameter, (2) Data augmentation, (3) Deeper model architecture, (4) Change the beta schedule, (5) Others**

You must **specify the strategies** you have used and report your **final evaluation results** (including the FID score and AFD rate). Please note that the discussion section constitutes an extra 10% of the total score.

Part 2: Optical Illusion Generation (25%)

Learning Objectives: Why do we need to learn this ?

Application

1. Learn How to Utilize Open-Source Models (e.g., from HuggingFace)
2. Exploring Training-Free Generative Methods Using Existing Models
3. Implement the Idea from Latest Top Conference Paper (e.g., CVPR)

It gives you a new idea to explore this area without more training or focusing only on performance.

Part 2-1: Prompt Design (5%)

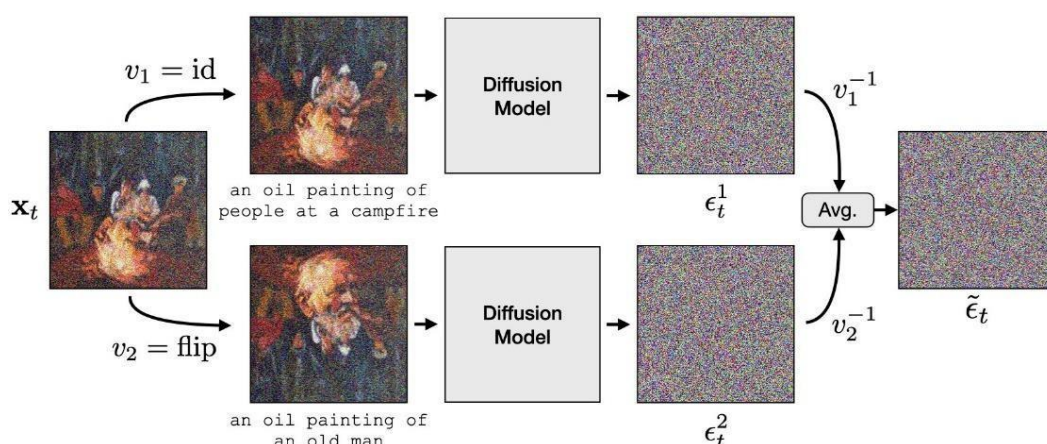
- Before generating an image, you must first determine its content using a text prompt. You need to design **two distinct prompts** for different perspectives. Your grade will be based on **creativity and diversity**. For example, if your prompts are “A photo of a wooden house” and “A photo of an upside-down wooden house,” they may not be different enough to earn a full score. Additionally, your prompts must be unique and original. If they are identical to someone else's, you will receive **0 points** for this section.

Part 2-2: Viewing Transformation (5%)

- We need to generate an optical illusion image that looks different through different views; therefore, the image requires a transformation to represent the view. Here, you are asked to complete **IdentityView** and **Rotate180View**. Both of them contain **view()** and **inverse_view()** that return the image after applying the transformation.

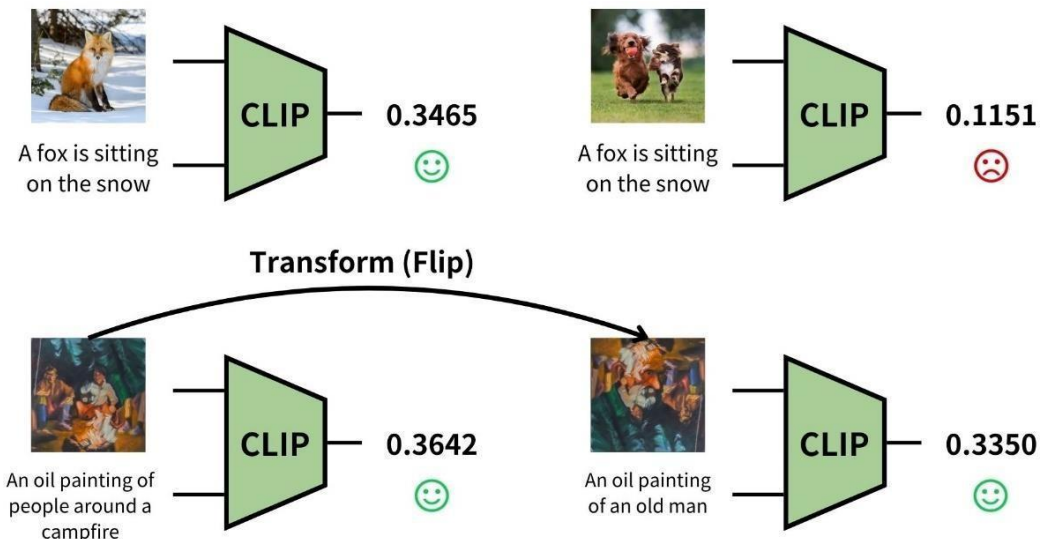
Part 2-3: Denoising Operation (10%)

- Unlike vanilla diffusion, the image should be processed through each **viewing function** separately, with the diffusion model predicting noise for each view independently. The predicted noises are then applied to their corresponding **inverse viewing functions**. Finally, the noise is removed by **averaging** the predicted noises. In the sample code, only the first prompt is considered for denoising. The goal is to implement these operations based on the original code to generate optical illusion images.



Part 2-4: Evaluation Baseline (5%)

- To evaluate the performance of your optical illusion image, we use the **CLIP score** to measure the similarity between the text and the image. A higher score indicates that the image better represents the text content. In this case, each text-image pair has its own CLIP scores, both of which must **exceed 0.3** to receive full points.



Report (30%)

- A report is required, which **will affect** your implementation score.
- The report should be written in **English**.
- Please save the report as a **.pdf** file. (font-size: 12)
- Display your results in the **report template in detail**.
- Please answer the questions/discussion as follows:

Part 1: Result & Discussion (20%)

- Discuss any observed issues (e.g., over-smoothing, artifacts).
- Explain which training strategies you used (e.g., data augmentation, deeper model, beta schedule changes, etc.).
- Provide experimental results (e.g., evaluation score, visual comparisons) to demonstrate the effectiveness of the modifications.

Part 2: Result & Discussion(10%)

- Discuss how prompt design strategies (e.g., descriptive vs. abstract prompts, inclusion of object-level cues, specificity, length) affect the result.
- Include failure cases: discuss examples where the visual anagram failed to represent the intended word or concept, and analyze why the failure occurred.

QA Page

If you have any questions about this homework, please ask them on the following E3 forum page. We will answer them as soon as possible. Additionally, we encourage you to answer other students' questions if you can.

Submission

Please prepare your source code, results, and report (.pdf) in `STUDENTID_hw5.zip`, following the file structure. Do **not** include an extra folder named `{student_id}_hw5` in the zip file.

 `{student_id}_hw5.zip`
└─  `diffusion.ipynb`
└─  `multi-view.ipynb`
└─  `report.pdf`

e.g. `112550999_hw5.zip`

Wrong submission format leads to -10 points.

In addition, this assignment requires you to submit your **training results**, including:

- **Model checkpoints**
- **Generated image results**

Please include the download link to these files in Part 3 of your report, following the format specified in the report template.

 Important reminders:

- You must upload the files to **Google Drive** only, and ensure that the download link is **publicly accessible**.
 - If the link is unavailable or restricted at the time of grading, **no excuses will be accepted**, and you may receive no points for that part.
- You must **strictly follow the report template structure**.
 - TAs will grade your submission based on the structure in the template.
 - If any required section is missing, points will be deducted accordingly.

Late Submission Policy

Late submissions will not be accepted.

Reference

- [1] [Machine Learning Material from Hung-yi Lee](#)
- [2] [Denoising Diffusion Probabilistic Models \(NeurIPS 2020\)](#)
- [3] [Denoising diffusion implicit models \(ICLR 2021\)](#)
- [4] [DeepFloyd IF on Hugging Face](#)
- [5] [Visual Anagrams \(CVPR 2024 oral\)](#)

Appendix

Sample from Gaussian Distribution

when we have such equation: $p(x) = \mathcal{N}(x; \mu, \sigma^2 I)$

it can be inferred like this: $x = \mu + \sigma * \epsilon$ where $\epsilon \sim \mathcal{N}(0, 1)$
random noise in code

Fréchet Inception Distance (FID Score)

